



Department of Artificial Intelligence and Multimedia Gaming

Lab # 06	Application of ICT (Fall-2025)
Course Title:	Application of ICT
Lab Instructor:	Rashida Naz
Credits (Theory + Lab):	3 Credit Hours (2 Hrs Theory and 1 Hr Lab)
Lab Duration:	2 Hrs
Lab Title:	Working with Excel Part 2

Objectives of the Lab:

In this lab, the students are going to learn:

- Concatenation
- Count Function
- If condition

COUNTBLANK Function

The **COUNTBLANK** function is a premade function in Excel, which counts blank cells in a range.

It is typed **=COUNTBLANK**

Note: The **COUNTBLANK** function is helpful to find empty cells in a range.

How to use the **=COUNTBLANK** function:

1. Select a cell
2. Type **=COUNTBLANK**
3. Double click the **COUNTBLANK** command
4. Select a range
5. Hit enter

Task # 1

Open Lab06.xlsx file, in the excel sheet namely: task1, Apply the **=COUNTBLANK** function to range **C2:C21**, to find the Pokemons which do not have a 2nd Type:

As follows:

	A	B	C	D	E	F	G	H	I
1	Name	Type 1	Type 2	Total stats					
2	Mankey	Fighting		305					
3	Poliwrath	Water	Fighting	510					
4	Victreebel	Grass	Poison	490					
5	Tentacool	Water	Poison	335					
6	Magneton	Electric	Steel	465					
7	Dewgong	Water	Ice	475					
8	Cloyster	Water	Ice	525					
9	Onix	Rock	Ground	385					
10	Dragonair	Dragon		420					
11	Pidgeotto	Normal	Flying	349					
12	Rattata	Normal		253					
13	Beedrill	Bug	Poison	395					
14	Doduo	Normal	Flying	310					
15	Kingler	Water		475					
16	Nidoqueen	Poison	Ground	505					
17	Hitmonchan	Fighting		455					
18	Charmeleon	Fire		405					
19	Arbok	Poison		438					
20	Gastly	Ghost	Poison	310					
21	Magikarp	Water		200					
22									
23			=COUNTBLANK						
24			 COUNTBLANK	Counts the number of empty cells in a specified range of cells					
25									

COUNTBLANK function, step by step:

1. Select C23
2. Write =COUNTBLANK
3. Double click **COUNTBLANK** in the menu
4. Select range C2:C21
5. Hit enter

COUNTA Function

The **COUNTA** function is a premade function in Excel, which counts all cells in a range that has values, both numbers and letters.

It is typed =COUNTA

How to use the =COUNTA function:

1. Select a cell
2. Type =COUNTA
3. Double click the **COUNTA** command

4. Select a range
5. Hit enter

Task 2

Open Lab06.xlsx file, in the excel sheet namely: task2, Apply the **=COUNTA** function to range **A2:A21**, counting Pokemons by their names, which are letters only:

	A	B	C	D	E
1	Name	Type 1	Type 2	Total stats	
2	Mankey	Fighting		305	
3	Poliwrath	Water	Fighting	510	
4	Victreebel	Grass	Poison	490	
5	Tentacool	Water	Poison	335	
6	Magneton	Electric	Steel	465	
7	Dewgong	Water	Ice	475	
8	Cloyster	Water	Ice	525	
9	Onix	Rock	Ground	385	
10	Dragonair	Dragon		420	
11	Pidgeotto	Normal	Flying	349	
12	Rattata	Normal		253	
13	Beedrill	Bug	Poison	395	
14	Doduo	Normal	Flying	310	
15	Kingler	Water		475	
16	Nidoqueen	Poison	Ground	505	
17	Hitmonchan	Fighting		455	
18	Charmeleon	Fire		405	
19	Arbok	Poison		438	
20	Gastly	Ghost	Poison	310	
21	Magikarp	Water		200	
22					
23	=COUNTA				
24	COUNTA	Counts the number of cells in a range that are not empty			
25	DCOUNTA				
26					

COUNTA function, step by step:

1. Select A23
2. Type =COUNTA
3. Double click **COUNTA** in the menu
4. Select range A2:A21
5. Hit enter

CONCAT Function

To concatenate is to link something together.

CONCAT is a function in Excel and is short for concatenate.

The **CONCAT** function is used to link multiple cells without adding any delimiters between the combined cell values.

It is typed =CONCAT

=CONCAT(**cell1**, **delimiter**, **cell2**)

Note: Delimiters are spacing or symbols used to separate content elements apart from each other.

Example of delimiters

comma , semicolon ; quotes " or ' braces {} pipes | slashes / \

Info: CONCAT is a new version of the old CONCATENATE function. Recent versions of Excel do not accept the old CONCATENATE function as it is only compatible with earlier versions of Excel. To use COCATENATE in the latest versions of Excel, use CONCAT.

How to Use CONCAT Function

To combine values from multiple Excel cells, use **CONCAT**.

Note: The different parts of the function are separated by a symbol, like comma , or semicolon ;

The symbol depends on your [Language Settings](#).

Step 1) Start the CONCAT function

1. Select a cell E2
2. Type =CONCAT
3. Double click the **CONCAT** command

	A	B	C	D	E	F	G
1	Name	Type 1	Total stats				
2	Mankey	Fighting	305		=CONCAT		
3	Poliwrath	Water	510				
4	Victreebel	Grass	490				
5	Tentacool	Water	335				
6	Magnetron	Electric	465				
7	Dewgong	Water	475				
8	Cloyster	Water	525				
9	Onix	Rock	385				
10	Dragonair	Dragon	420				
11	Pidgeotto	Normal	349				
12	Rattata	Normal	253				
13	Beedrill	Bug	395				
14	Doduo	Normal	310				
15	Kingler	Water	475				
16	Nidoqueen	Poison	505				
17	Hitmonchan	Fighting	455				
18	Charmeleon	Fire	405				
19	Arbok	Poison	438				
20	Gastly	Ghost	310				
21	Magikarp	Water	200				

Step 2) Link cells

4. Select a cell (A2)
5. Add a comma (,)
6. Add a space inside of quotation marks to create a delimiter (" ")
7. Select another cell (A3)
8. Hit enter

Note: The **CONCAT** function does not provide any delimiters between cell values used in the function. If you want your cell values to be displayed with delimiters, you have to add the delimiters between the selected cells. You can use any symbol as a demiliter, if you want the delimiter to be a space, enter " ".

	A	B	C	D	E
1	Name	Type 1	Total stats		
2	Mankey	Fighting	305		=CONCAT(A2," ",A3)
3	Poliwrath	Water	510		
4	Victreebel	Grass	490		
5	Tentacool	Water	335		
6	Magnetron	Electric	465		
7	Dewgong	Water	475		
8	Cloyster	Water	525		
9	Onix	Rock	385		
10	Dragonair	Dragon	420		
11	Pidgeotto	Normal	349		
12	Rattata	Normal	253		
13	Beedrill	Bug	395		
14	Doduo	Normal	310		
15	Kingler	Water	475		
16	Nidoqueen	Poison	505		
17	Hitmonchan	Fighting	455		
18	Charmeleon	Fire	405		
19	Arbok	Poison	438		
20	Gastly	Ghost	310		
21	Magikarp	Water	200		
22					

The function returns the content of A2 and A3 separated with a " " dilimiter.

IF Function

The **IF** function is a premade function in Excel, which returns values based on a **true** or **false condition**.

It is typed =IF and has 3 parts:

=IF(logical_test, [value_if_true], [value_if_false])

The **condition** is referred to as logical_test, which can check things like:

- If a number is **greater than** another number >
- If a number is **smaller than** another number <
- If a number or text is **equal** to something =

Note: You can decide both the return values and the condition.

Note: The different parts of the function are separated by a symbol, like comma , or semicolon ;

The symbol depends on your [Language Settings](#).

Example IF function (equal to)

Check if the Pokemon type is grass:

The condition is if the "Type 1" value for the Pokemon is "Grass".

The function returns "Yes" or "No".

	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318		
3	Ivysaur	Grass	405		
4	Venusaur	Grass	525		
5	Charmander	Fire	309		
6	Charmeleon	Fire	405		
7	Charizard	Fire	534		
8	Squirtle	Water	314		
9	Wartortle	Water	405		
10	Blastoise	Water	530		
11					

Example **IF** function, step by step:

1. Select the cell D2
2. Type =IF
3. Double click the **IF** command

	A	B	C	D	E	F	G	H	I	J	K
1	Name	Type 1	Total	Grass Type							
2	Bulbasaur	Grass	318	=IF							
3	Ivysaur	Grass	405	IF	Checks whether a condition is met, and returns one value if TRUE, and another value if FALSE						
4	Venusaur	Grass	525	IFERROR							
5	Charmander	Fire	309	IFNA							
6	Charmeleon	Fire	405	IFS							
7	Charizard	Fire	534								
8	Squirtle	Water	314								
9	Wartortle	Water	405								
10	Blastoise	Water	530								
11											

4. Specify the condition B2="Grass"
5. Type ,

6. Specify the value "Yes" for when the condition is **TRUE**
7. Type ,
8. Specify the value "No" for when the condition is **FALSE**
9. Hit enter

	A	B	C	D	E	F	G
1	Name	Type 1	Total	Grass Type			
2	Bulbasaur	Grass	318	=IF(B2="Grass"; "Yes"; "No")			
3	Ivysaur	Grass	405	IF (logical_test; [value_if_true]; [value_if_false])			
4	Venusaur	Grass	525				
5	Charmander	Fire	309				
6	Charmeleon	Fire	405				
7	Charizard	Fire	534				
8	Squirtle	Water	314				
9	Wartortle	Water	405				
10	Blastoise	Water	530				
11							

Since the value in cell B2 is "Grass", the condition is **true** and the function will return "Yes".

Note: Text values needs to be in quotes: " "

The function can be repeated with the [filling function](#) for each row to perform the same check for each Pokemon:

	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318	=IF(B2="Grass"; "Yes"; "No")	
3	Ivysaur	Grass	405	=IF(B3="Grass"; "Yes"; "No")	
4	Venusaur	Grass	525	=IF(B4="Grass"; "Yes"; "No")	
5	Charmander	Fire	309	=IF(B5="Grass"; "Yes"; "No")	
6	Charmeleon	Fire	405	=IF(B6="Grass"; "Yes"; "No")	
7	Charizard	Fire	534	=IF(B7="Grass"; "Yes"; "No")	
8	Squirtle	Water	314	=IF(B8="Grass"; "Yes"; "No")	
9	Wartortle	Water	405	=IF(B9="Grass"; "Yes"; "No")	
10	Blastoise	Water	530	=IF(B10="Grass"; "Yes"; "No")	
11				IF (logical_test; [value_if_true]; [value_if_false])	
12					

Now, each row has a check for Grass Type:

	A	B	C	D	E
1	Name	Type 1	Total	Grass Type	
2	Bulbasaur	Grass	318	Yes	
3	Ivysaur	Grass	405	Yes	
4	Venusaur	Grass	525	Yes	
5	Charmander	Fire	309	No	
6	Charmeleon	Fire	405	No	
7	Charizard	Fire	534	No	
8	Squirtle	Water	314	No	
9	Wartortle	Water	405	No	
10	Blastoise	Water	530	No	
11					

Example IF Function (greater than)

Check if the Pokemon's total stats is **greater than** 500:

The function returns "Yes" or "No".

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318		
3	Ivysaur	Grass	405		
4	Venusaur	Grass	525		
5	Charmander	Fire	309		
6	Charmeleon	Fire	405		
7	Charizard	Fire	534		
8	Squirtle	Water	314		
9	Wartortle	Water	405		
10	Blastoise	Water	530		
11					

Example **IF** function, step by step:

1. Select the cell D2
2. Type =IF
3. Double click the **IF** command

	A	B	C	D	E	F	G	H	I
1	Name	Type 1	Total	More than 500 Total Stats					
2	Bulbasaur	Grass	318	=IF					
3	Ivysaur	Grass	405	IF	Checks whether a condition is met, and returns one value if TRUE, and another value if FALSE				
4	Venusaur	Grass	525	IFERROR					
5	Charmander	Fire	309	IFNA					
6	Charmeleon	Fire	405	IFS					
7	Charizard	Fire	534						
8	Squirtle	Water	314						
9	Wartortle	Water	405						
10	Blastoise	Water	530						
11									

4. Specify the condition $C2 > 500$
5. Type ,
6. Specify the value "Yes" for when the condition is **TRUE**
7. Type ,
8. Specify the value "No" for when the condition is **FALSE**
9. Hit enter

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318	=IF(C2>500; "Yes"; "No")	
3	Ivysaur	Grass	405	IF (logical_test; [value_if_true]; [value_if_false])	
4	Venusaur	Grass	525		
5	Charmander	Fire	309		
6	Charmeleon	Fire	405		
7	Charizard	Fire	534		
8	Squirtle	Water	314		
9	Wartortle	Water	405		
10	Blastoise	Water	530		
11					

Since the value in cell C2 is 318, the condition is **false** and the function will return "No".

The function can be repeated with the [filling function](#) for each row to perform the same check for each Pokemon:

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318	=IF(C2>500; "Yes"; "No")	
3	Ivysaur	Grass	405	=IF(C3>500; "Yes"; "No")	
4	Venusaur	Grass	525	=IF(C4>500; "Yes"; "No")	
5	Charmander	Fire	309	=IF(C5>500; "Yes"; "No")	
6	Charmeleon	Fire	405	=IF(C6>500; "Yes"; "No")	
7	Charizard	Fire	534	=IF(C7>500; "Yes"; "No")	
8	Squirtle	Water	314	=IF(C8>500; "Yes"; "No")	
9	Wartortle	Water	405	=IF(C9>500; "Yes"; "No")	
10	Blastoise	Water	530	=IF(C10>500; "Yes"; "No")	
11				IF (logical_test; [value_if_true]; [value_if_false])	
12					

Now, each row has a check for having more than 500 total stats:

	A	B	C	D	E
1	Name	Type 1	Total	More than 500 Total Stats	
2	Bulbasaur	Grass	318	No	
3	Ivysaur	Grass	405	No	
4	Venusaur	Grass	525	Yes	
5	Charmander	Fire	309	No	
6	Charmeleon	Fire	405	No	
7	Charizard	Fire	534	Yes	
8	Squirtle	Water	314	No	
9	Wartortle	Water	405	No	
10	Blastoise	Water	530	Yes	
11					

Task 3

Apply Concatenation on Task3 sheet in the Lab6.xlsx, Concatenate two columns First name and Last name.

Task 4

Open Task4 in the lab6.xlsx file, Apply if condition to check the cost of product based on the following criteria:

If Cost< 5000: display too Low, otherwise display normal

Task 5

Open Task4 in the lab6.xlsx file, Apply conditional formatting on the column cost. Highlight the values with red if the cost is less than 5000

Task 6

Add a S# column in the table given in the Task4 in the lab6.xlsx file

Task 7

Apply Filtering on all four sheets given in the lab6.xlsx file.